Normalizing Flows Question Bank: 40 Verified Exam and Quiz Items with Worked Solutions

There are enough authoritative sources to build a 20–30-item quiz that matches slides 72–105, but I found no officially published flow exam problem with solutions. So this bank adapts the canonical lecture examples (Stanford CS236) and the primary papers, and pairs them with 40 newly constructed, hand-verified exercises that test transfer rather than recall.

TL;DR

- The lecture deck matches Stanford CS236 Lecture 7/8 almost line for line: X=4Z gives 1/8, X=exp(Z) gives 1/(2x), NICE/Real-NVP, MAF/IAF, Parallel WaveNet, MintNet, Gaussianization flows. The easy questions below therefore quote that source directly. The medium and hard items use new numbers, maps and scenarios, and every answer was recomputed.
- Public flow exams with official solutions are scarce. The Berkeley CS294-158-SP20 course site lists "HW2: Flow Models (out 2/12, due 2/25) [solutions]", but its tasks are coding notebooks ("Autoregressive Flow on 2D Data; RealNVP on 2D Data; Autoregressive Flows for Images; RealNVP on Higher Dimensions"), and the CS236 midterm solutions only circulate as unofficial uploads. The harder items are therefore original derivations grounded in the primary papers: Glow, MAF, Parallel WaveNet, Gaussianization Flows, MintNet, Rezende & Mohamed, and the Papamakarios et al. JMLR review.
- Suggested quiz: 8 easy + 10 medium + 8 hard items, chosen from the 40 below (the recommended picks are listed at the end). The strongest "unseen-case" items are the planar-flow collision (H3), the Gaussian multi-information/rotation calculation (H6), the KL-direction reasoning for Parallel WaveNet (H8) and the single-layer MAF failure (H9).

Key Findings

- **Lecture alignment.** CS236 Lecture 8 (Ermon) contains the exact statements in the deck. Examples: "x, z need to be continuous and have the same dimension"; github "Computing the determinant for an n × n matrix is $O(n^3)$"; MAF "Sampling is sequential and slow… O(n) time"; IAF "Fast to sample from, slow to evaluate likelihoods of data points"; Parallel WaveNet "Improves sampling efficiency over original Wavenet… by 1000x" (the paper's abstract adds it runs "more than 20 times faster than real-time, and is deployed online by Google Assistant", with MOS 4.41 ± 0.078 matching the teacher's 4.41 ± 0.069). MintNet: "If all the diagonal elements of the Jacobian matrix are (strictly) positive, the transformation is invertible". Gaussianization: "Even though each dimension is marginally Gaussian, they are not jointly Gaussian… copula model (Sklar, 1959)".

- **Primary-paper facts usable as answer keys:**
  - Glow: log-det of a 1×1 convolution = $h \cdot w \cdot \log|\det W|$. Det cost drops from $O(c^3)$ to $O(c)$ via the LU parameterization. arxiv ActNorm log-det = $h \cdot w \cdot \text{sum}(\log|s|)$. neurips
  - Planar flow (Rezende & Mohamed 2015): with h=tanh, "a sufficient condition for f(z) to be invertible is that $w^{\intercal}u \geqslant -1$". mlr
  - MAF vs IAF (Papamakarios et al. 2017): "MAF is more suitable for density estimation, because each example requires only one pass through the model whereas IAF requires D". ed
  - Parallel WaveNet: loss $D_{KL}(P_S \| P_T) = H(P_S, P_T) - H(P_S)$. The entropy term "prevents the student's distribution from collapsing to the mode of the teacher". arxiv
  - Gaussianization Flows: $D_{KL}(p(x) \| N(0,I)) = I(x) + J_m(x)$. Dimension-wise Gaussianization decreases J_m and leaves $I(x)$ invariant, and a rotation leaves the total unchanged. arxiv
  - MintNet: a lower-triangular-Jacobian map with a non-singular Jacobian is invertible. The inverse is computed by a fixed-point iteration that is locally convergent for step size $0<\alpha<2$. arxiv
- **What is NOT available.** Berkeley CS294-158 HW2 "Flow Models" (SP19 lists "HW2.PDF, HW2_template.tex, hw2_q2.pkl"; SP20 lists coding tasks with "[solutions]") asks students to implement autoregressive flows and RealNVP on 2D data and images. It contains no written derivation problems. The CS236 HW2 (Fall 2018) Problem 1 is a VAE problem. coursehero A "CS236 Mid Term Soln" (130 points, 7 questions incl. flows) exists only as an unofficial Scribd upload that I could not access. scribd Do not cite it.

Details: The Question Bank

Each item lists: format · difficulty · question · answer · worked solution · source URL. Notation: φ is the standard normal pdf, Φ the CDF, log = natural log, $\frac{1}{2}\log(2\pi) = 0.918939$, $\log(2\pi) = 1.837877$.

EASY (recall and understanding of the lecture)

E1 · Single-choice · Easy Which statement best motivates normalizing flows, given autoregressive (AR) models and VAEs? A) AR models have intractable likelihoods; VAEs have tractable ones. B) AR models give tractable likelihoods but no direct feature learning; VAEs learn latent features but have intractable marginal likelihood; flows aim for a latent-variable model with tractable likelihood. C) Flows are needed because VAEs cannot be sampled. D) Flows compress data to a lower-dimensional latent. **Answer:** B. **Solution:** The CS236 notes say: "Autoregressive models provide tractable likelihoods but no direct mechanism for learning features, whereas variational autoencoders can learn feature representations but have

intractable marginal likelihoods." github D is false because flows require equal dimensions. Source: https://deepgenerativemodels.github.io/notes/flow/

E2 · True/False · Easy "In a normalizing flow, x and z must be continuous and have the same dimension." **Answer:** True. **Solution:** The map must be a bijection (in fact a diffeomorphism) between R^n and R^n. The change-of-variables formula needs a square Jacobian, because a determinant exists only for square matrices. jmlr github Source: https://deepgenerativemodels.github.io/assets/slides/cs236_lecture8.pdf

E3 · Numeric fill-in · Easy (lecture example) Z ~ U[0,2] and X = 4Z. What is p_X(4)? **Answer:** 1/8 = 0.125. **Solution:** h(x) = x/4, so |h'| = 1/4 and p_X(4) = p_Z(1)·1/4 = (1/2)(1/4) = 1/8. As a sanity check, X ~ U[0,8]. The tempting wrong answer 1/2 comes from substituting into the pdf without the derivative factor. github Source: https://deepgenerativemodels.github.io//assets/slides/cs236_lecture7.pdf

E4 · Single-choice · Easy For a flow x = f(z), which expression is correct? A) $p_X(x) = p_Z(f(x))\cdot|\det \partial f/\partial z|$ B) $p_X(x) = p_Z(f^{-1}(x))\cdot|\det \partial f^{-1}(x)/\partial x| = p_Z(z)\cdot|\det \partial f(z)/\partial z|^{-1}$ C) $p_X(x) = p_Z(f^{-1}(x))$ D) $p_X(x) = p_Z(z)\cdot\det \partial f/\partial z$ (no absolute value) **Answer:** B. **Solution:** This follows from probability conservation, p(x)dx = p(z)dz, together with $\det(A^{-1}) = \det(A)^{-1}$. github The absolute value is needed because orientation-reversing maps have negative determinants. Source: https://lilianweng.github.io/posts/2018-10-13-flow-models/

E5 · Multiple-select · Easy Which operations does a trained flow support without an extra inference network? (Select all.) A) Exact log-likelihood via x→z B) Sampling via z→x C) Latent inference $z = f^{-1}(x)$ D) A lower bound (ELBO) only **Answer:** A, B, C. **Solution:** Lecture 8 lists "Exact likelihood evaluation via inverse transformation $x \mapsto z$", "Sampling via forward transformation $z \mapsto x$", and "Latent representations inferred via inverse transformation (no inference network required!)". github The ELBO belongs to VAEs. Source: https://deepgenerativemodels.github.io/assets/slides/cs236_lecture8.pdf

E6 · Matching · Easy Match each model to its training objective. Models: (1) GAN, (2) VAE, (3) Flow, (4) Autoregressive. Objectives: (a) exact log-likelihood via change of variables; (b) adversarial minimax game; (c) evidence lower bound; (d) exact log-likelihood via the chain rule of conditionals, which needs an ordering. **Answer:** 1-b, 2-c, 3-a, 4-d. **Solution:** Weng: "VAE inexplicitly optimizes the log-likelihood… by maximizing the ELBO"; a flow's "loss function is simply the negative log-likelihood". AR models factorize p(x)=∏p(x_i|x_<i). github Source: https://lilianweng.github.io/posts/2018-10-13-flow-models/

E7 · True/False · Easy "The NICE additive coupling layer x_{d+1:n} = z_{d+1:n} + m_θ(z_{1:d}) is volume preserving even when m_θ is a deep, non-invertible neural network." **Answer:** True. **Solution:** The Jacobian is lower block-triangular with identity blocks on the diagonal, so det = 1. github The inverse is z_{d+1:n} = x_{d+1:n} − m_θ(x_{1:d}). github Neither the inverse nor

the determinant needs m_θ to be invertible. Source: https://deepgenerativemodels.github.io/assets/slides/cs236_lecture8.pdf

E8 · Short text fill-in · Easy "Computing the determinant of a general n×n Jacobian costs O(**), whereas for a triangular Jacobian it costs O(**)." **Answer:** n^3; n. **Solution:** For a triangular matrix the determinant is the product of the diagonal entries. Lecture 8 states this directly. github github Source: https://deepgenerativemodels.github.io/assets/slides/cs236_lecture8.pdf

E9 · Matching · Easy Match each model to its computational profile. (1) MAF, (2) IAF. (a) Fast, parallel likelihood of external data; slow, sequential sampling. (b) Fast, parallel sampling; slow likelihood of external data; fast likelihood of its own samples (cached z). **Answer:** 1-a, 2-b. **Solution:** "MAF: Fast likelihood evaluation, slow sampling; IAF: Fast sampling, slow likelihood evaluation… github Note: Fast to evaluate likelihoods of a generated point (cache z1,…,zn)." github Source: https://deepgenerativemodels.github.io/assets/slides/cs236_lecture8.pdf

E10 · True/False · Easy "Glow replaces Real-NVP's fixed channel-reversal permutation with a learned invertible 1×1 convolution, which generalizes any channel permutation." **Answer:** True. **Solution:** Glow: "a 1×1 convolution with equal number of input and output channels is a generalization of a permutation operation". W is initialized as a random rotation, so its log-det is 0 at initialization. arxiv Source: https://arxiv.org/pdf/1807.03039

MEDIUM (new numbers, new maps, applied understanding)

M1 · Numeric · Medium Z ~ U(0,1) and X = 3Z + 2. What is the support of X, and what is p_X(3.5)? **Answer:** Support [2,5]; p_X(3.5) = 1/3 ≈ 0.3333. **Solution:** h(x) = (x−2)/3, so |h'| = 1/3. p_Z(h(3.5)) = p_Z(0.5) = 1, and therefore p_X = 1/3. Source (method): https://blog.evjang.com/2018/01/nf1.html

M2 · Numeric · Medium Z ~ U(0,1) and X = −ln Z. Derive p_X and evaluate p_X(ln 2). **Answer:** p_X(x) = e^{−x} for x > 0, i.e. Exponential(1). p_X(ln 2) = 0.5. **Solution:** f(z) = −ln z is monotone decreasing on (0,1). The inverse is h(x) = e^{−x}, so |h'(x)| = e^{−x}, and p_Z = 1 on (0,1) gives p_X(x) = e^{−x}. At x = ln 2 this is e^{−ln 2} = 1/2. The derivative here is negative, which is why the absolute value matters. Source (method): https://deepgenerativemodels.github.io//assets/slides/cs236_lecture7.pdf

M3 · Numeric · Medium Z ~ N(0,1) and X = 2Z + 1. Compute log p_X(1). **Answer:** −1.6121. **Solution:** z = (1−1)/2 = 0, so log p_Z(0) = −0.918939. Then log|dz/dx| = log(1/2) = −0.693147, and the total is −1.612086. As a check, X ~ N(1, 4) and its peak density is 1/(2√(2π)) = 0.19947, whose log is −1.6121. Source (method): https://lilianweng.github.io/posts/2018-10-13-flow-models/

M4 · Numeric · Medium $Z \sim U(0,1)$ and $X = Z^2$. Compute $p_X(0.64)$. What value does the "common mistake" (substitution without the derivative) give? **Answer:** 0.625; the mistake gives 1. **Solution:** $h(x) = \sqrt{x}$ and $h'(x) = 1/(2\sqrt{x})$. At $x = 0.64$, $h' = 1/(2 \cdot 0.8) = 0.625$, so $p_X(0.64) = p_Z(0.8) \cdot 0.625 = 0.625$. The full pdf $1/(2\sqrt{x})$ integrates to 1 on (0,1). The mistake $p_Z(\sqrt{0.64}) = 1$ does not. Source (method): https://deepgenerativemodels.github.io//assets/slides/cs236_lecture7.pdf

M5 · Numeric · Medium (extends lecture example) $Z \sim U[0,2]$ and $X = \exp(Z)$, so $p_X(x) = 1/(2x)$ on $[1, e^2]$. Compute (a) $p_X(2)$ and (b) $P(X \leqslant e)$. **Answer:** (a) 0.25; (b) 0.5. **Solution:** (a) $1/(2 \cdot 2) = 1/4$. (b) Either compute $\int_1^e 1/(2x)dx = \frac{1}{2} \cdot \ln e = \frac{1}{2}$, or note that $P(X \leqslant e) = P(Z \leqslant 1) = \frac{1}{2}$. The second route shows that probabilities of events are preserved under the bijection, while densities are not. Source: https://deepgenerativemodels.github.io//assets/slides/cs236_lecture7.pdf

M6 · Numeric · Medium An affine flow $x = Az + b$ uses $A = [[2,1],[0,3]]$, $b = (1,-1)$ and $z \sim N(0, I_2)$. Compute $\log p_X(x)$ at $x = (4, 2)$. **Answer:** −4.6296. **Solution:**

1. Solve $Az = x - b = (3, 3)$. The second row gives $3z_2 = 3$, so $z_2 = 1$. The first row gives $2z_1 + z_2 = 3$, so $z_1 = 1$. Hence $z = (1,1)$.
2. $\log N(z; 0, I) = -\log(2\pi) - \frac{1}{2}(1+1) = -1.837877 - 1 = -2.837877$.
3. $\det A = 2 \cdot 3 - 1 \cdot 0 = 6$, and $\log 6 = 1.791759$.
4. $\log p_X = -2.837877 - 1.791759 =$ **−4.629636**. Source (method): https://jmlr.org/papers/volume22/19-1028/19-1028.pdf

M7 · Numeric · Medium A 2D NICE additive coupling layer is $x_1 = z_1$, $x_2 = z_2 + m(z_1)$ with $m(z_1) = z_1^2$, and $z \sim N(0, I)$. For $x = (1, 3)$, find z, $\det J$ and $\log p_X(x)$. **Answer:** $z = (1, 2)$; $\det J = 1$; $\log p_X = -4.3379$. **Solution:** $z_1 = 1$ and $z_2 = 3 - 1^2 = 2$. $J = [[1,0],[2z_1,1]]$, so $\det J = 1$. $\log p_X = -1.837877 - \frac{1}{2}(1 + 4) = -4.337877$. Because the layer preserves volume, the density equals the base density at z. Source (method): https://deepgenerativemodels.github.io/assets/slides/cs236_lecture8.pdf

M8 · Numeric · Medium A Real-NVP forward pass has $n = 4$ and $d = 2$ with $z = (1, -1, 0.5, 2)$. The conditioners are $\alpha(z_1,z_2) = (0.5z_1, z_2)$ and $\mu(z_1,z_2) = (z_1z_2, 1)$. Compute x and $\log|\det J|$. Does the layer expand or contract volume here? **Answer:** $x = (1, -1, -0.1756, 1.7358)$; $\log|\det J| = -0.5$; it contracts ($\det = 0.6065$). **Solution:** $\alpha = (0.5, -1)$ and $\mu = (-1, 1)$.

- $x_3 = 0.5 \cdot e^{0.5} - 1 = 0.5 \cdot 1.648721 - 1 = -0.175639$.
- $x_4 = 2 \cdot e^{-1} + 1 = 0.735759 + 1 = 1.735759$.
- $\log \det = \Sigma\alpha = 0.5 - 1 = -0.5$, so $\det = e^{-0.5} = 0.606531$, which is less than 1. Source: https://deepgenerativemodels.github.io/assets/slides/cs236_lecture8.pdf

M9 · Numeric · Medium A Real-NVP inverse with likelihood has n = 2 and d = 1, base N(0,I), $\alpha(u) = 0.5u$ and $\mu(u) = u$. For x = (2, 5), compute z and log p_X(x). **Answer:** z = (2, 1.1036); log p_X = −5.4469. **Solution:**

1. $z_1 = 2$, $\alpha = 1$ and $\mu = 2$, so $z_2 = (5 - 2)e^{\{-1\}} = 1.103638$.
2. $\log|\det \partial f^{-1}/\partial x| = -\alpha = -1$.
3. $\log N(2) = -0.918939 - 2 = -2.918939$.
4. $\log N(1.103638) = -0.918939 - 1.218017/2 = -1.527948$.
5. Total: −2.918939 − 1.527948 − 1 = **−5.446887**. Source: https://deepgenerativemodels.github.io/assets/slides/cs236_lecture8.pdf

M10 · Numeric · Medium A three-layer flow has |det J| equal to 2, 0.5 and 3 at the visited points, and $\log p_Z(z_0) = -2$. What is log p_X(x)? **Answer:** −3.0986. **Solution:** Log-dets add: ln 2 + ln 0.5 + ln 3 = ln 3 = 1.098612. Then $\log p_X = \log p_Z - \Sigma \log|\det J_{f_k}| = -2 - 1.098612 = -3.098612$. The determinant of a product is the product of the determinants. jmlr github Source: https://jmlr.org/papers/volume22/19-1028/19-1028.pdf

M11 · Numeric · Medium A Glow invertible 1×1 convolution acts on a 16×16×c activation, and det W = 2. What does it contribute to the log-determinant? What does the LU parameterization change? **Answer:** 256·ln 2 = 177.446 nats. The LU form makes $\log|\det W| = \Sigma\log|s|$, which costs O(c) instead of $O(c^3)$. arxiv **Solution:** The same c×c matrix is applied at each of the h·w = 256 spatial positions. The Jacobian is therefore block-diagonal with 256 copies of W, giving log-det = $h \cdot w \cdot \log|\det W|$ github = 256 × 0.693147 = 177.4457. Source: https://arxiv.org/pdf/1807.03039

M12 · Numeric · Medium An ActNorm layer on an 8×8×3 tensor has per-channel scales s = (2, 0.5, 4). What is its log-det? Is it volume preserving? **Answer:** 64·ln 4 = 88.723; not volume preserving. **Solution:** Glow gives log-det = $h \cdot w \cdot \Sigma\log|s_i|$ neurips = 64·(ln2 + ln0.5 + ln4) = 64·ln 4 = 64 × 1.386294 = 88.7228. **Follow-up trap:** with s = (2, 0.5, 1) the log-det is 0, so the layer is volume preserving even though no individual scale equals 1. Source: https://arxiv.org/pdf/1807.03039

M13 · Numeric · Medium A one-layer 2D MAF uses $\mu_1 = \alpha_1 = 0$, $\mu_2(x_1) = x_1$ and $\alpha_2(x_1) = 0.5x_1$, with base N(0,I). For x = (2, 3), compute z and log p_X(x). How many sequential network passes are needed? **Answer:** z = (2, 0.3679); log p_X = −4.9055; one pass, because all μ_i and α_i come from x in parallel. **Solution:**

1. $z_1 = (2 - 0)/e^0 = 2$, and $z_2 = (3 - 2)/e^1 = 0.367879$.
2. $\log|\det \partial z/\partial x| = -(\alpha_1 + \alpha_2) = -1$.
3. $\log N(2) = -2.918939$, and $\log N(0.367879) = -0.918939 - 0.067668 = -0.986607$.
4. Total: −4.905546. Source: https://deepgenerativemodels.github.io/assets/slides/cs236_lecture8.pdf

M14 · Numeric · Medium A one-layer 2D IAF uses $\mu_1 = \alpha_1 = 0$, $\mu_2(z_1) = 2z_1$ and $\alpha_2(z_1) = z_1$. Sample $z = (1, 0.5)$. Compute x. Then: given only an external x, how many sequential steps are needed to recover z in D dimensions? **Answer:** x = (1, 3.3591); D sequential steps. **Solution:** $x_1 = 1$ and $x_2 = e^1 \cdot 0.5 + 2 = 1.359141 + 2 = 3.359141$, all computed in one pass from z. To invert, z_2 needs $\mu_2(z_1)$ and so needs z_1 first. In general z_i needs z_<i, which gives D sequential steps. Source: https://arxiv.org/pdf/1705.07057

M15 · Multiple-select · Medium After step 1 of a Gaussianization flow (dimension-wise $z^{(i)} = \Phi^{-1}(F_i(x^{(i)}))$), which statements hold? (Select all.) A) Each coordinate is marginally N(0,1). B) The vector is jointly N(0,I). C) The Jacobian is diagonal. D) The multi-information I(x) is unchanged. E) J_m (the sum of marginal KLs to N(0,1)) is driven to 0, assuming the marginal CDFs are exact. **Answer:** A, C, D, E. **Solution:** CS236: "Even though each dimension is marginally Gaussian, they are not jointly Gaussian". github Meng et al.: dimension-wise Gaussianization "will decrease J_m(x) and leave I(x) invariant". The map acts elementwise, so its Jacobian is diagonal. arxiv Sources: https://deepgenerativemodels.github.io/assets/slides/cs236_lecture8.pdf ; https://arxiv.org/pdf/2003.01941

M16 · Numeric · Medium Gaussianize X ~ Exponential(1) with $z = \Phi^{-1}(1 - e^{-x})$. Compute z for x = ln 2 and for x = ln 4. **Answer:** z = 0 and z = 0.6745. **Solution:** $F(\ln 2) = 1 - \frac{1}{2} = 0.5$, so $z = \Phi^{-1}(0.5) = 0$. $F(\ln 4) = 1 - \frac{1}{4} = 0.75$, so $z = \Phi^{-1}(0.75) = 0.674490$. **Extension:** $dz/dx = p_X(x)/\varphi(z) = 0.25/\varphi(0.6745)$. Here $\varphi(0.6745) = 0.398942 \cdot e^{-0.227468} = 0.31778$, so $dz/dx \approx 0.787$. Source (method): https://deepgenerativemodels.github.io/assets/slides/cs236_lecture8.pdf

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HARD (abstraction, deduction, unseen cases)

H1 · Open-ended / numeric · Hard Why does a ReLU layer x = ReLU(Wz) break a flow, even with W invertible? Propose a fix. Then compute the fixed layer's log|det| for a 5-dimensional input with 2 negative pre-activations, using leaky slope 0.1 and W = I. **Answer:** ReLU is not injective, because every negative input maps to 0. The Jacobian has zero diagonal entries there, so det = 0, log-det = −∞, and probability mass collapses onto a lower-dimensional set, which leaves the density undefined. Fix: leaky ReLU with slope a > 0, which is strictly monotone. $\log|\det| = 2 \cdot \ln 0.1 = -4.6052$. **Solution:** Flows need a bijection with a non-singular Jacobian. Leaky ReLU has derivative 1 or a on each coordinate, so log|det| = (number of negatives)·ln a = 2·(−2.302585) = −4.605170. Source (principles): https://jmlr.org/papers/volume22/19-1028/19-1028.pdf

H2 · Open-ended · Hard What goes wrong if Real-NVP used $x_{d+1:n} = z_{d+1:n} \odot \alpha_\theta(z_{1:d}) + \mu_\theta$ instead of $\odot \exp(\alpha_\theta)$? **Answer:** Invertibility and log-likelihood both break wherever an α entry equals 0. A continuous network whose output changes sign must pass through 0, so there the map is not invertible, det = 0 and log|det| = −∞. Near 0, log|α| gives exploding gradients and numerical instability. Exponentiating forces the scale to be strictly

positive (invertible everywhere), keeps orientation consistent, and makes log-det = Σα linear in the network output. **Solution:** Papamakarios et al.: for an affine transformer "Invertibility is guaranteed if $\alpha_i \neq 0$, and this can be easily achieved by e.g. taking $\alpha_i = \exp \tilde{\alpha}_i$… the log absolute Jacobian determinant is $\Sigma\tilde{\alpha}_i$." jmlr Source: https://jmlr.org/papers/volume22/19-1028/19-1028.pdf

H3 · Numeric + deduction · Hard A planar flow $f(z) = z + u\cdot\tanh(w^Tz + b)$ in 2D has $w = (1,1)$ and $b = 0$. (a) With $u = (1,0)$, compute $f(z)$ and $\det J$ at $z = (1,0)$. (b) With $u = (-3,0)$, show that the map is not invertible by finding two distinct points with the same image. **Answer:** (a) $f = (1.7616, 0)$, $\det = 1.4200$. (b) $z_a = (0,0)$ and $z_b \approx (2.985, 0)$ both map to $(0,0)$. **Solution:**

- (a) $w^Tz = 1$ and $\tanh 1 = 0.761594$, so $f = (1.761594, 0)$. Then $\det = 1 + h'(1)\cdot u^Tw = 1 + (1 - 0.580026)\cdot 1 = 1.419974$.
- (b) Now $u^Tw = -3 < -1$, which violates the sufficient condition $w^Tu \geqslant -1$. mlr At $s = w^Tz = 0$, $\det = 1 - 3\cdot 1 = -2$. For large $|s|$, $\det \to 1$, so the determinant changes sign.
- Collision: we need $z + u\cdot\tanh(s) = 0$ with $z = (t, 0)$, so $t = 3\tanh(t)$. The root $t^* \approx 2.985$ checks out: $3\cdot\tanh(2.985) = 3\cdot 0.99491 \approx 2.985$. So $z_b = (2.985, 0)$ maps to $(0,0)$, the same image as $z_a = (0,0)$. Sources: https://proceedings.mlr.press/v37/rezende15-supp.pdf ; https://deepgenerativemodels.github.io//assets/slides/cs236_lecture7.pdf

H4 · Open-ended / design · Hard Your data has odd dimension $n = 5$. Design coupling layers so that every dimension is transformed and the determinant stays $O(n)$. **Answer (reference):**

- Any split $1 \leqslant d < n$ is valid; the lecture says "for any $1 \leqslant d < n$". github Use $d = 2$ (or 3): layer 1 keeps $x_{1:2}$ fixed and transforms $x_{3:5}$ conditioned on them, and layer 2 swaps roles, with $d = 3$ kept fixed ($x_{3:5}$) and $x_{1:2}$ transformed.
- Alternatively, insert a permutation or learned 1×1 (n×n) invertible linear layer between couplings. A permutation has $|\det| = 1$, and an LU-parameterized linear layer has $O(n)$ det.
- At least two coupling layers are needed before every dimension has been transformed. Each layer's Jacobian is block-triangular with diagonal $\exp(\alpha)$ entries (or 1s), so the det is still a product of diagonals. **Solution:** Odd dimension only makes the two blocks unequal in size. It does not affect invertibility or the triangular structure. Sources: https://deepgenerativemodels.github.io/assets/slides/cs236_lecture8.pdf ; https://lilianweng.github.io/posts/2018-10-13-flow-models/

H5 · Multiple-select · Hard Which of these maps $R^2 \to R^2$ are valid flow layers (diffeomorphisms with tractable det)? (Select all.) A) $x = Az$ with $A = [[1,2],[2,4]]$ B) $x_1 = z_1$, $x_2 = z_2\cdot\exp(\tanh(z_1)) + z_1^3$ C) $x = z + 0.5\cdot\tanh(z)$ (elementwise) D) $x_1 = z_1 + z_2$, $x_2 = z_1 + z_2$ E) $x = \mathrm{ReLU}(z)$ F) $x = Qz$ with Q a 2×2 rotation **Answer:** B, C, F. **Solution:**

- A: $\det = 4 - 4 = 0$, so singular.

- B: an affine coupling with positive scale; det = $\exp(\tanh z_1) > 0$.
- C: the derivative $1 + 0.5 \cdot \operatorname{sech}^2(z)$ lies in $[1, 1.5]$, always > 0.
- D: rank 1.
- E: not injective.
- F: $|\det Q| = 1$, and the map is volume preserving. Source (principles): https://jmlr.org/papers/volume22/19-1028/19-1028.pdf

H6 · Numeric · Hard (Gaussianization, unseen case) Let $x \sim N(0, \Sigma)$ with $\Sigma = [[1, 0.6],[0.6, 1]]$. (a) Compute $D_{KL}(p(x) \| N(0,I))$, the multi-information $I(x)$ and $J_m(x)$. (b) Rotate by 45° (the eigenvectors of Σ). Recompute I and J_m. (c) What does the next marginal-Gaussianization step do? **Answer:**

- (a) KL = 0.2231, I = 0.2231, $J_m = 0$.
- (b) I = 0, $J_m = 0.0650 + 0.1581 = 0.2231$.
- (c) It rescales each rotated coordinate by $1/\sqrt{\lambda}$, which drives J_m to 0. The total KL becomes 0 and the data is fully Gaussianized. **Solution:**
- (a) Both marginals are N(0,1), so $J_m = 0$. Then $KL = \frac{1}{2}(\operatorname{tr}\Sigma - 2 - \ln\det\Sigma) = \frac{1}{2}(0 - \ln 0.64) = \frac{1}{2}(0.446287) = 0.223144$, so $I = 0.223144$. This matches the bivariate formula $-\frac{1}{2}\ln(1-\rho^2)$.
- (b) The eigenvalues are 1.6 and 0.4. The coordinates become independent, so I = 0. Each marginal contributes $KL(N(0,\lambda) \| N(0,1)) = \frac{1}{2}(\lambda - 1 - \ln\lambda)$: for $\lambda = 1.6$, $\frac{1}{2}(0.6 - 0.470004) = 0.064998$; for $\lambda = 0.4$, $\frac{1}{2}(-0.6 + 0.916291) = 0.158145$. The sum is 0.223143. The total is unchanged, as the paper states for rotations, because "N(0,I) is rotationally invariant". arxiv Source: https://arxiv.org/pdf/2003.01941

H7 · True/False · Hard "Fitting a flow by maximum likelihood (forward KL on x-space) is equivalent to minimizing the reverse KL between the distribution of f^{-1}(data) and the base N(0,I) in z-space." **Answer:** True. Strictly, it is the KL from the pushed-forward data distribution to the base: $D_{KL}(p_x \| p_x) = D_{KL}(p_u \| p_u)$. **Solution:** The JMLR review proves this KL equality (their Eq. 21), jmlr and it underlies the lecture's Gaussianization argument: data passed through f^{-1} should look Gaussian. arxiv The 5-layer MAF demo checks exactly this, since samples of $u = f^{-1}(x)$ look Gaussian. Sources: https://jmlr.org/papers/volume22/19-1028/19-1028.pdf ; https://deepgenerativemodels.github.io/assets/slides/cs236_lecture8.pdf

H8 · Open-ended · Hard (KL direction) Parallel WaveNet trains the IAF student with $D_{KL}(s \| t) = E_{x\sim s}[\log s(x) - \log t(x)]$. Why not $D_{KL}(t \| s)$? **Answer:**

1. Computability. $D_{KL}(s \| t)$ needs (i) samples from the student, which is fast because the IAF is parallel; (ii) the student density of its own samples, which is fast via the cached z; and (iii) the teacher density of those samples, which is fast because a WaveNet/MAF scores in parallel. $D_{KL}(t \| s)$ would need samples from the teacher, which are sequential: the paper measures the autoregressive teacher at "172 timesteps/second" versus the

student's "over 500,000 timesteps/second" on an NVIDIA P100. It would also need the student density at teacher samples, which are external points that an IAF can only score with D sequential steps.

2. Behavior. Reverse KL is mode-seeking. The −H(P_S) entropy term "prevents the student's distribution from collapsing to the mode of the teacher". arxiv **Solution:** The paper defines D_KL(P_S‖P_T) = H(P_S,P_T) − H(P_S). It explains that direct MLE training of the IAF "would be impractical, as the inference procedure required to estimate the log-likelihoods is sequential and slow". arxiv Sources: https://arxiv.org/pdf/1711.10433 ; https://deepgenerativemodels.github.io/assets/slides/cs236_lecture8.pdf

H9 · Open-ended / deduction · Hard In the lecture, a 1-layer MAF fails to fit a 2D cross-shaped density but a 5-layer MAF succeeds. Explain the failure in terms of the model's conditionals. What does stacking with different orderings add? **Answer:** With a Gaussian base and an affine transformer, one layer gives $x_1 \sim N(\mu_1, e^{\{2\alpha_1\}})$, a single Gaussian, and $x_2|x_1$ is Gaussian. A cross-shaped density has a non-Gaussian, multimodal-looking marginal for x_1, and the model cannot represent it. Stacking layers lets later layers warp the earlier output nonlinearly. Changing the ordering lets each variable be the "conditioned-on" one in some layer. **Solution:** Papamakarios et al.: "a single affine autoregressive transformation of a multivariate Gaussian results in a distribution whose conditionals… will necessarily be Gaussian… expressive flows can still be obtained by stacking multiple affine autoregressive layers." jmlr Sources: https://jmlr.org/papers/volume22/19-1028/19-1028.pdf ; https://arxiv.org/pdf/1705.07057

H10 · Matching / design · Hard Pick MAF, IAF or MAF→IAF distillation for each application. (1) Anomaly detection: score millions of logged sensor vectors by density. (2) Real-time text-to-speech at 24 kHz. (3) A flexible variational posterior q(z|x) inside a VAE. (4) Train on a large audio dataset, then deploy with fast generation. **Answer:** 1-MAF, 2-IAF (trained by distillation), 3-IAF, 4-MAF teacher → IAF student (Parallel WaveNet). **Solution:**

- (1) needs one-pass density of external data.
- (2) needs parallel sampling.
- (3) VI only needs samples from q plus the density of those samples, which is exactly what IAF does cheaply. This is why IAF was introduced for variational inference. harvard arxiv
- (4) MLE is cheap for the teacher, and the student learns via D_KL(s‖t). github Sources: https://arxiv.org/pdf/1705.07057 ; https://deepgenerativemodels.github.io/assets/slides/cs236_lecture8.pdf

H11 · Numeric · Hard Parallel WaveNet stacks IAF layers with the same ordering, $x^i = x^{i-1} \cdot s^i + \mu^i$ (the paper's student "typically consisted of 4 flows with 10, 10, 10, 30 layers respectively"). Two layers have $(s^1, \mu^1) = (2, 1)$ and $(s^2, \mu^2) = (3, -1)$. Compute s_tot, μ_tot and the log-det contribution per timestep. **Answer:** s_tot = 6, μ_tot = 2, log-det = ln 6 = 1.7918. **Solution:** $x^1 = 2z$

+ 1, and $x^2 = 3(2z + 1) - 1 = 6z + 2$. This matches the paper's formulas $s_{tot} = \prod s^i$ and $\mu_{tot} = \Sigma_i \mu^i \prod_{j>i} s^j$ arxiv $= 1 \cdot 3 + (-1) = 2$. Source: https://arxiv.org/pdf/1711.10433

H12 · Numeric / design · Hard (Murphy elementwise flow) The elementwise map $h(u) = au + b + c/(1 + (du + e)^2)$ has $d > 0$. Derive the condition on a, c, d for h to be strictly increasing. Is (a, c, d) = (1, 1, 1) valid? Is (0.5, 1, 1)? **Answer:** $a > (9/(8\sqrt{3})) \cdot |c| \cdot d \approx 0.6495 \cdot |c| \cdot d$. The triple (1,1,1) is valid; (0.5,1,1) is not. **Solution:** Let $t = du + e$. Then $h' = a - c \cdot d \cdot 2t/(1+t^2)^2$. The magnitude of $g(t) = 2t/(1+t^2)^2$ peaks where $1 - 3t^2 = 0$, i.e. $t = 1/\sqrt{3}$, giving $g = (2/\sqrt{3})(9/16) = 9/(8\sqrt{3}) = 0.649519$. Requiring $h' > 0$ for all u gives $a > 0.6495|c|d$. Then $1 > 0.6495$ holds, and $0.5 < 0.6495$ fails. Source (layer form, Murphy PML2 ch. 23 / Papamakarios review): https://jmlr.org/papers/volume22/19-1028/19-1028.pdf

H13 · Open-ended · Hard (dimensionality) A colleague proposes a flow from a 16-dimensional Gaussian latent to 784-pixel MNIST images "to get a compressed representation". Evaluate the proposal and suggest alternatives. **Answer:**

- The proposal is invalid. A diffeomorphism needs equal dimensions, and a 16→784 map can only cover a 16-dimensional manifold, so its density in R^{784} is degenerate and the Jacobian is non-square.
- Flows give a lossless, same-dimensional latent: generation is deterministic given z, with no compression.
- Alternatives: (a) a VAE (or VAE with a flow posterior/prior); (b) an autoencoder plus a flow on its latent; (c) a multi-scale Real-NVP/Glow that factors out variables early. That last option still has 784 total latent dimensions, but they are hierarchically organized. **Solution:** The "same dimension" requirement appears explicitly in Lecture 8 and in the JMLR definition, github which says diffeomorphisms "require that u be D-dimensional as well". jmlr Sources: https://jmlr.org/papers/volume22/19-1028/19-1028.pdf ; https://deepgenerativemodels.github.io/notes/flow/

H14 · Multiple-select · Hard (MintNet) Which statements about MintNet are correct? (Select all.) A) Masked (PixelCNN-like) convolutions impose an ordering, so the Jacobian is triangular. B) A strictly positive Jacobian diagonal guarantees invertibility. C) The inverse is computed in closed form in one pass. D) The inverse is computed by a parallelizable fixed-point iteration that is locally convergent for step size $0<\alpha<2$. E) Masking reduces the effective receptive field compared with an unmasked CNN. **Answer:** A, B, D, E. **Solution:** Song et al.: "If $f \in \mathcal{F}$ [lower-triangular Jacobian] and $J_f(x)$ is non-singular for all x… then f is invertible." Algorithm 1 iterates $x_t \leftarrow x_{t-1} - \alpha \cdot \mathrm{diag}(J_f)^{-1}(f(x_{t-1}) - z)$, and "Theorem 2. The iterative method… is locally convergent whenever $0 < \alpha < 2$." The paper also notes that x^3 is invertible even though its Jacobian is singular at 0, so the condition is sufficient, not necessary. arxiv Sources: https://arxiv.org/pdf/1907.07945 ; https://deepgenerativemodels.github.io/assets/slides/cs236_lecture8.pdf

Additional items (fill gaps or substitute)

A1 · Numeric · Medium. $Z \sim N(0,1)$ and $X = Z^3$. Compute $p_X(8)$. **Answer:** $\varphi(2)/12 = 0.053991/12 = 0.004499$. **Solution:** $h(x) = x^{1/3}$ and $h'(x) = (1/3)x^{-2/3}$. At $x = 8$, $h' = (1/3)(1/4) = 1/12$. Follow-up: at $x = 0$ the density blows up, because $f'(0) = 0$ and so f^{-1} is not differentiable there. This is why flows require diffeomorphisms, as MintNet's x^3 example shows. arxiv Source: https://arxiv.org/pdf/1907.07945

A2 · Numeric · Easy–Medium. For CIFAR-10 ($n = 32 \cdot 32 \cdot 3 = 3072$), by what factor does a triangular Jacobian reduce determinant cost versus a dense one? **Answer:** $n^2 = 9{,}437{,}184$ (about 10^7). Source: https://deepgenerativemodels.github.io/assets/slides/cs236_lecture8.pdf

A3 · Ordering · Medium. Put the Parallel WaveNet pipeline in order: (a) use the student at test time; (b) train the teacher by MLE; (c) sample x from the student and score it under student and teacher; (d) update the student to minimize $D_{KL}(s\|t)$. **Answer:** b → c → d → a. Source: https://deepgenerativemodels.github.io/assets/slides/cs236_lecture8.pdf

A4 · Ordering · Medium. Order one Glow step of flow (forward direction). **Answer:** ActNorm → invertible 1×1 conv → affine coupling. readthedocs Source: https://arxiv.org/pdf/1807.03039

A5 · Multiple-select · Medium. Which transformations are volume preserving? (i) a NICE additive coupling; (ii) a channel permutation; (iii) a Householder reflection $I - 2vv^T/\|v\|^2$; (iv) a Real-NVP coupling with $\Sigma\alpha = 0$ at a point; (v) a NICE rescaling with $s = (2, 2)$. **Answer:** i, ii, iii are always volume preserving. iv is locally volume preserving at that point only. v is not (det = 4). Sources: https://deepgenerativemodels.github.io/assets/slides/cs236_lecture8.pdf ; https://arxiv.org/pdf/2003.01941

A6 · Multiple-select · Medium (Neural Spline Flows). Which are true of rational-quadratic spline transformers? (a) analytic inverse; (b) monotonic because the knot derivatives are positive; (c) identity outside $[-B, B]$; (d) require bisection to invert. **Answer:** a, b, c. **Solution:** Durkan et al. propose "monotonic rational-quadratic splines… retaining analytic invertibility". arxiv Evaluation finds the bin by binary search, but the inverse within a bin is closed-form. arxiv-vanity jmlr Source: https://arxiv.org/pdf/1906.04032

A7 · Short fill-in · Easy. In a coupling flow $x^A = f(u^A; \Theta(u^B))$, $x^B = u^B$, $\det J(f) = \det J(____)$. **Answer:** f, the coupling function applied to the transformed block. arxiv Source: https://arxiv.org/html/1908.09257v4

A8 · True/False · Medium. "For an IAF, evaluating the density of a point it generated itself costs as much as for an external data point." **Answer:** False. Caching z makes it a single pass, which is the key enabler of Parallel WaveNet. github github Source: https://deepgenerativemodels.github.io/notes/flow/

A9 · Numeric · Medium. Gaussianize U ~ U(0,1) via $z = \Phi^{-1}(u)$. What is z at u = 0.975, and what is |dz/du| there? **Answer:** z = 1.960. |dz/du| = 1/φ(1.96) = 1/0.058441 = 17.11. **Solution:** φ(1.96) = 0.398942·e^{−1.9208} = 0.398942 × 0.146491 = 0.058441. Source (method): https://deepgenerativemodels.github.io/assets/slides/cs236_lecture8.pdf

A10 · True/False · Medium. "MAF is a generalization of Real-NVP: a coupling layer is a special case of an autoregressive transformation." **Answer:** True. MAF's abstract says MAF "is closely related to Inverse Autoregressive Flow and is a generalization of Real NVP". arxiv Source: https://arxiv.org/abs/1705.07057

Recommendations

Suggested 26-question quiz (easy → hard):

- **Easy (8):** E1, E2, E3, E4, E6, E7, E9, A7
- **Medium (10):** M2, M4, M6, M7, M9, M10, M11, M13, M15, A3
- **Hard (8):** H1, H2, H3, H6, H8, H9, H10, H13

Format coverage in that set: single-choice (E1, E4), multiple-select (E5-type via M15), true/false (E2, E7), matching (E6, E9, H10), ordering (A3), numeric (M-series, H3, H6), short fill-in (A7) and open-ended (H1, H2, H8, H9, H13).

For the HTML build:

- Give numeric items a tolerance of ±0.01 absolute (or ±0.5% relative) and show the worked solution on submit.
- For log-likelihood items, state "natural log" explicitly in the question stem. Many students default to $\log_2$ when they think "bits".
- Turn H3 and H6 into two-step items: compute first, then a "what does this imply" choice. This is where transfer is actually tested.
- Add a distractor for the classic mistake to every 1D change-of-variables item (E3, M1, M4, M5). Distractor values: 1/2 for E3, 1 for M1, 1 for M4.

Caveats

- **No official flow exam with solutions was retrievable.** Berkeley CS294-158 lists "HW2: Flow Models… [solutions]", but the material is coding notebooks hosted off-site. google The CS236 midterm solutions found online are an unofficial Scribd upload and should not be used as a source. scribd Every M/H item here is original and grounded in the cited paper or slide. The "source URL" therefore points to the concept's authority, not to a published copy of the exercise.
- **Slide attribution quirks.** The CS236 slides label planar flows "(Rezende & Mohamed, 2016)", but the paper is ICML 2015. arxiv + 2 The slides call the Parallel WaveNet teacher a

"MAF", while the paper's teacher is an autoregressive WaveNet. github github The two are equivalent under the flow view, but the wording differs.

- **Speed-up figures.** The Parallel WaveNet paper reports "more than 20 times faster than real-time" and "more than 1000× speed-up relative to original WaveNet". arxiv alphaXiv's overview of arXiv:1711.10433 claims "a speedup of more than 2,900 times"; this traces to the paper's own benchmark (172 timesteps/second for the autoregressive teacher vs over 500,000 timesteps/second for the distilled model, ≈2,907×), but the paper's stated headline is "more than 1000×", so use 1000× in the quiz.
- **Paraphrases vs quotes.** Slide statements are quoted from the CS236 PDF text extraction, which can lose formatting. The Gaussianization (Eq. 3–5) and MintNet (Theorem 1/2, Algorithm 1) quotes come from the arXiv PDFs. The H12 invertibility bound $(9/(8\sqrt{3}))$ is my own derivation, checked by calculus, not quoted from Murphy.
- **Numerical rounding.** All values are given to 4–6 significant figures. $\Phi^{-1}(0.75) = 0.674490$ and φ values come from standard tables. Re-verify in code (scipy.stats.norm) before publishing the answer key.